

Lamp (cap lamp, LED, neon lantern, rainproof torch)

Produit : Electronic / Measuring devices

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CHINA

Product's definition

Category

In which category the product is classified ?

Products covered: lamp (cap lamp, LED, neon lantern, rainproof torch...).

GB/T 39008-2020 double-ended integrated LED lamp performance requirements

Definition

What is the exact definition of the product ?

And if it exists, what is the interpretation of the authorities regarding it ?

Definition of electrical and electronic products

Electrical and electronic products means the equipment and supporting products that work by relying on electric currents or electromagnetic fields or are manufactured for the purpose of generating, transmission and measuring electric currents and electromagnetic fields, with the rated working voltage of not more than 1,500 volts measured in direct current or not more than 1,000 volts measured in alternating current, excluding the equipment involving electrical energy production, transmission and distribution. (Administrative Measures for Restricted Use of the Hazardous Substances Contained in Electrical and Electronic Products)

Accordingly, most of the products involved in this section basically meet the definition of electrical and electronic products.

Definition of mechanical and electrical products

The term “mechanical and electrical products” refers to the mechanical equipment, electric equipment, transport tools, electronic products, electrical appliances, apparatus, instruments, metal products, and the parts and components and elements thereof. (Article 2 of Administrative Measures for Import of Mechanical and Electrical Products)

The specific scope of mechanical and electrical products is stipulated in the appendix of the “Administrative Measures for Import of Mechanical and Electrical Products”, which covers a wide range, including electrical and electronic products, instruments and meters, various luminaries and lighting devices, clocks and watches and their parts.

Among them, the import of some mechanical and electrical products is subject to automatic licensing management. The specific product scope is stipulated in the “Automatic Import License Mechanical and Electrical Products Catalog”.

Labelling

Language

In which language must the information on the product or on the packaging be indicated? If English is sufficient please indicate NA.

Must use Chinese language .

Mandatory labelling on product

What are the mandatory labelling on the product ?

General requirements

General Requirements of Labeling

1	All necessary information concerning the proper and safe use and (or) related service and maintenance of the products shall be included
2	Clear and easy to be identified, the smallest English letter shall not be less than 1.5 mm
3	Must use Chinese language
4	The contents shall be the same with that contained in other printed materials (such as advertisement and package)

Content of Labeling and Instruction for Use

1	Product summarizes (product name, variety, specifications, model, the main purpose and working conditions, etc.)
2	Precautions for safe use
3	Structural characteristics and working principle
4	Technical characteristics
5	Size, weight
6	Installation and adjustment

7	Use and operation
8	Malfunction analysis and exclusion
9	Safety protection device and incident handling
10	Maintenance and repair
11	Transport and storage
12	Unpacking and inspection
13	Environmental or other requirements
14	Graphs, charts, photographs (eg outline drawings, installation drawings, schematics, etc.)

Marks on lamps

Marks on Lamps

Type A	Type B	Type C
1) Rated wattage	1)Origin marking	1)The object which was covered
2) Special light source	2)Rated voltage	2)Harsh environments
3)Cold-beam	3)Ambient temperature	
4) Bowl-shaped mirror reflector bulb	4) Class II/III luminaire	
5)Protective cover	5) IP number	
6)Trigger warning	6)Model	
7)Self-shielded bulb	7)Symbol	
	8) End	
	9)Interconnected lamps	

Type A: the mark shall be visible in the outer surface of lamp or the reverse side of its cover removed when changing the light source.

Type B: the mark shall be visible in the outer surface of lamp or the reverse side of its cover or its componentes removed when installing.

Type C: the mark shall be visible when the lamp is installed for normal use and the light source is in position.

Labeling Requirement of Lamp

Item	Content
Box	Name and address of the manufacturer
	Name and model of product
	Product number in the box
	Standard of implementation
	Others

Certificate

1) Name and registered address of the manufacturer

2) Inspection date

3) Inspector signature

Instruction for use

Basic parameters, rated value, etc. of the lamp

Applicable occasions for the lamp

Nominal illumination and distance

Composition of the product and circuit diagram

Installation methods and precautions

The difference between three types of marks: A, B and C

- According to Article 3.2 of GB7000.1—2007 Luminaires - Part 1: General requirements and tests, labeling of type A/B/C should be marked on lamps clearly and durably. The three types are divided in accordance with the timing the labeling shall be visible. So the positions of the labeling are different correspondingly.
- Type A includes: rated wattage, special light source, cold-beam, bowl-shaped mirror reflector bulb, protective cover, trigger warning, self-shielded bulb. This kind of labeling shall be visible in the outer surface (except the fitting surface) of lamp or the reverse side of its cover removed when changing the light source.
- Type B includes: origin marking, rated voltage, ambient temperature, classâ...j/â...¢ luminaire, IP number, model, symbol, end, interconnected lamps. This kind of labeling shall be visible in the outer surface of lamp or the reverse side of its cover or its components removed when installing.
- Type C includes: the illuminated objects, harsh environments. This kind of labeling shall be visible when the lamp is installed for normal use and the light source is in position.
- Pursuant to Article 1.2.9 of GB7000.1—2007 Luminaires - Part 1: General requirements and tests, portable general purpose luminaires are luminaires that could be move from one position to another after connecting the power when normally used.
- Subject to the Article 5 of GB 7000.204—2008 Luminaires - Part 2-4: Particular requirements - Portable general purpose luminaires, portable general purpose luminaires should also comply with GB7000.1—2007 Luminaires - Part 1: General requirements and tests stated above. And thus the lamps you refer to should be marked with these three types of labeling simultaneously.

Labeling requirement for cap lamp

In accordance with GB 7957-2003 (General requirements for safety of cap lamp)

General Requirements

1

The mark should be clear, perpetual and corrosion-resistant

2

In the conspicuous place of outer part of cap lamp, the symbol of ExS1, national standard code and “anti-explosion certification” code should be labeled

Common Requirements of the cap lamp and Outer Package

1	Trademark
2	Date of manufacture
3	Model
4	Name of manufacturer

Outer package

1	Standard number implemented
2	Name of the product
3	Address of manufacturer

GB/T36949—2018 Double (Capped LED Lamps Designed to Retrofit Liner Fluorescent Lamps) Performance Specifications

3.9.1 Label requirements

The minimum packaging of the lamp or the contents of the instructions on the instruction manual shall comply with the requirements of Table 5.

Table 5 Minimum packaging of double-ended LED lights, lamps or marking requirements on the instruction manual

Project	Lamp body	Minimum packaging or instruction manual for the lamp
Replacement of double-ended fluorescent lamp type (tube diameter and power) meter with ballast type (electronic or inductive)	-	×
Product name	-	×
Model	×	×
Date of manufacture (year, season or month, year/month by number, quarter by roman; or production lot that can be traced back to production date)	×	-
Applicable power supply voltage / ambient temperature	×	×
Rated luminous flux / lm	-	×
Light distribution type	-	×
Rated power / w	-	×

Power factor		×
General color rendering index Ra	-	×
Correlated color temperature / k	-	×
Rated life / h	-	×
Lamp type	-	×
Execution standard or number	-	×

Note: “×” is the mandatory content; “-” is the content of the selection

a Replacement double-ended LED lamp containing a fluorescent lamp electronic control device luminaire. The model number of the control device is described in the manual.

Regulatory requirements for

product labeling

1. The producers and importers of electrical and electronic products shall, in accordance with the national or industrial standards for the marking of the restricted use of the hazardous substances contained in electrical and electronic products, mark the hazardous substances contained in the electrical and electronic products they put on markets, and indicate the names and contents of hazardous substances or whether the elements or products which contain such substances may be recycled, the fact that improper use or disposal thereof may affect the environment or human health and other information. Where the aforesaid information cannot be marked on the products due to the restrictions in product volume, shape, surface materials or functions, it shall be indicated in the product manuals.
2. The producers and importers of electrical and electronic products shall, in accordance with the national or industrial standards for the marking of the restricted use of the hazardous substances contained in electrical and electronic products, indicate the EFUP on the electrical and electronic products they produce or import. Where the aforesaid information cannot be marked on the products due to the restrictions in product volume, shape, surface materials or functions, it shall be indicated in the product manuals.

Mandatory labelling on packaging

What are the mandatory labelling on the packaging ?

Labeling requirement for cap lamp (In accordance with GB 7957-2003 General requirements for safety of camp lamp)

General Requirements

1	The mark should be clear, perpetual and corrosion-resistant
2	In the conspicuous place of outer part of cap lamp, the symbol of ExS1, national standard code and “anti-explosion certification” code should be labeled

Common Requirements of the cap lamp and Outer Package

1	Trademark
2	Date of manufacture
3	model
4	Name of manufacturer

Outer package

1	Standard number implemented
2	Name of the product
3	Address of manufacturer

GB/T36949—2018 Double (Capped LED Lamps Designed to Retrofit Liner Fluorescent Lamps) Performance Specifications

3.9.2 Packaging requirements

The lights should be individually packaged and then packed in a box.

The number of lights in the box should be marked on the box.

The product should be accompanied by a product certificate or stamped with a stamp and stamped with the date of inspection.

GB/T 39008-2020 double-ended integrated LED lamp performance requirements

Packaging requirements

The lamp should be individually packaged and then assembled in a packing box. The number of lights in the box should be marked on the packing box. The packing box should be accompanied by a product certificate or a qualified seal, and the inspection date should be stamped.

Mandatory labelling on the instructions manual

What are the mandatory labelling on the instructions manual (if one is required) ?
In which format, do we have to deliver the instruction manual to our customers ?

Warnings - General ones

Are general warnings required ? If yes, which ones ?

Warnings - Specific ones

Are specific warnings required ? If yes which ones ?

Target Age

Is there any target age applicable? If so, does it have to be indicated and in which form?

Sanctions / Fines

What are the sanctions / fines in case of non conformity ?

Comments

Comments

Environment

Recycling - Eco taxes

Which regulations concerning ecodesign and recycling apply to this product? Are there any ecotaxes applicable to this category of product?

Regulatory requirements for electrical and electronic product

Regulatory requirements for

product design

The designers of electrical and electronic products may not violate the compulsory standards or the standards that must be implemented as prescribed by laws, administrative regulations and rules in the process of designing electrical and electronic products, and shall, on the premise of satisfying the technological requirements, adopt the schemes for making the said substances harmless, less harmful, easy to be degraded or convenient for being recycled, among others, in accordance with the national or industrial standards for the restricted use of the hazardous substances contained in electrical and electronic products,

Product production supervision requirements

- The manufacturers of electrical and electronic products may not violate the compulsory standards or the standards that must be implemented as prescribed by laws, administrative regulations and rules in the process of manufacturing electrical and electronic products, and shall, in accordance with the national or industrial standards for the restricted use of the hazardous substances contained in electrical and electronic products, adopt the materials, technologies and technological processes with higher resource utilization rates, easy to be recycled and disposed of and conducive to environmental protection, and restrict or eliminate the use of the hazardous substances in products.
- The manufacturers of electrical and electronic products may not allow the electrical and electronic products that fail to comply with the requirements of these Measures to leave factories or be sold.

regulatory requirements for packaging

The manufacturers and importers of electrical and electronic products may not violate the compulsory standards or the standards that must be implemented as prescribed by laws, administrative regulations and rules when making and using the packages of electrical and electronic products, and shall adopt the materials harmless, easy to be degraded and convenient for being recycled, and comply with the national or industrial standards for the use of packages.

Regulatory requirements stipulated in Administrative Measures for Prevention and Control of Environmental Pollution by Electronic Waste

Regulatory requirements for

product production

The producers of electrical and electronic products may in accordance with the national or industrial standards for the restricted use of the hazardous substances contained in electrical and electronic products, restrict or eliminate the use of the hazardous substances in products.

Regulatory requirements for information disclosure

Producers, importers and sellers of electronic and electrical products shall disclose the lead, mercury, cadmium, hexavalent chromium, polybrominated biphenyls (PBB), polybrominated diphenyl ethers (PBDE) and other toxic and hazardous substances contained in their products in accordance with relevant national regulations. As well as information on the impact of improper use or disposal on the environment and human health, and tips on how to use or dispose of the product in an environmentally sound manner after it is discarded.

Recycling and disposal regulatory requirements

- Producers, importers and sellers of electronic and electrical products shall establish a recycling system in accordance with relevant national regulations, recycle waste products, and be responsible for storage, utilization or disposal in an environmentally sound manner;
- If the producers, sellers and importers of electronic products discard electronic and electrical products, they shall provide or entrust the electronic wastes to the dismantling, utilization and disposal units (including individual industrial and commercial households) with corresponding business scope listed in the directory (including the temporary directory) for dismantling, utilization or disposal.

Labelling

What are the mandatory labellings to put on the product or on the packaging regarding the environmental rules ?

Technical rules / Standards

General safety requirements

What are the general safety requirements ? Does a risk assessment have to be conducted ? If yes, does it concern specific risk ?

The safety requirement of neon lamp:

The appearance, initial characteristics, average life and lux maintenance of neon lamp shall comply with the technical requirement of GB19261-2009 General and safety requirements of neon lamps.

Photoelectricity of Neon Lamp

Type	Color	Start voltage(Max)/VLamp voltage/V		Current/mA	Brightness (Min)×103cd/m2
Neon glow lamp	Red	1100+1200 L	230+700L~	25	2.0
			270+800L		
Mercury-argon glow lamp	Green	400+450 L	230+450L~		3.5
			270+650L		
Blue			1.4		
White			3.5		
Yellow			3.2		

The Life of Neon Lamp

No	Type	Average life/h	Lux maintenance/% (Lighted to 2000h)
1	Neon glow lamp	e10000	e95
2	Mercury-argon glow lamp	e8000	e90

GB/T 36949-2018 double-ended LED lamp (for replacing straight tube fluorescent lamp) performance requirements

The safety requirements of the lamp shall be in accordance with the requirements of IEC 62776.

GB/T 39075-2020 Safety requirements for organic light-emitting diode (OLED) panels for general lighting

requirements

General requirements: The design and production of OLED panels should ensure their safety in normal use and will not cause harm to users and the surrounding environment. Once it fails, the OLED panel should be able to fail safely.

The requirements and tests related to OLED panels in this standard are equally important.

General test requirements: Unless otherwise specified, the test should be carried out at an ambient temperature of 25 ±5 . Except for other provisions in this standard, the test current should be within ±1% of the rated current. The tested OLED panel should be installed according to the manufacturer's installation instructions. If there are multiple installation methods, the unfavorable installation method should be selected.

Logo

Content and location

The logo of the OLED panel should meet the requirements of Table 1.

Table 1 The content and location of the logo

parameter	product	Packaging, product data sheet or leaflet
Manufacturer or trademark		—
polarity		—

Model or production code	
Rated current or rated current range	—
Rated voltage or rated voltage range	—
rated power	—
	—
Power type	
	—
Shape or size	
	—
Connection information	
	—
Installation Notes	
	—
range of working temperature	
IP rating	—
Lamp design information	—

Note: — is the required content;-is the content of the selected table

The connection information should include the method of mechanical and electrical connection. The electrical connection information may include the type of drive.

For the IP grade mark, the protection grade symbol is required by Chapter 3 of IEC60598-1.

Durability and clarity of the logo

the mark should be clear and durable. Check its eligibility by visual inspection and the following methods: After lightly rubbing the mark with a smooth cloth soaked in water for 15 seconds, the mark should be clear.

structure

Overview: Wood, cotton, silk, paper and similar fibrous materials should not be used as insulators.

Check its eligibility by visual inspection.

Mechanical strength: The OLED panel should have sufficient mechanical strength. Use vibration test to check its eligibility. For the vibration test, the OLED panel should be installed in accordance with 4.2. The sinusoidal vibration test should be carried out in accordance with the requirements of GB/T2423.10-2019 with the following parameters: displacement: 0.35mm; acceleration: 50m/s; frequency range: 10hz~500hz; vibration axis: 3; duration: 3 times 10 cycle.

After the vibration test, the OLED panel should work for 15 minutes under the conditions of 4.2.

Eligibility: After the test, perform a visual inspection on the OLED panel to confirm its eligibility. Any breakage or fragments of glass, generation of fire, smoke or flammable gas, and loose parts that impair safety are not acceptable. The electrical contacts cannot be touched before the vibration test, nor can they become accessible after the test.

Internal short circuit

OLED panels with internal short circuits should not cause any harm.

Line channel

The line channel should be smooth and free of sharp edges and burrs that can damage the wire insulation. Parts, such as screws with sharp edges, should not protrude into the wiring channel.

Mechanical hazard

OLED panels with glass corners should not have sharp edges or corners that may cause danger during installation, normal operation or maintenance. OLED panels with metal foil plastic sheets should have protective materials to prevent sharp edges or corners from causing harm during installation, normal operation or maintenance.

Failure status

Failures that may occur during the expected use of the OLED panel should not impair its safety. Use the following overpower test to check its eligibility. Unless otherwise specified by the manufacturer or the responsible seller, the overpower test should be carried out at an ambient temperature of $25^{\circ}\text{C} \pm 5^{\circ}\text{C}$, and the ambient temperature should be maintained at $\pm 2^{\circ}\text{C}$ during the entire test.

The OLED panel should light up under the rated current. The input power should be monitored. During the test, the input power should be continuously increased until 150% of the rated power or rated current.

Insulation resistance and electric strength

Insulation resistance: in accordance with the requirements of 10.2.1 in IEC60598-1.

Electric strength: in accordance with the requirements of 10.2.2 in IEC60598-1.

Thermal stress: OLED panels should be able to withstand thermal stress. The thermal stress test shall be carried out at the specified temperature in an environmental test chamber. The temperature range is $60 \sim 70$. Choose the appropriate temperature. During the test, the temperature should be kept within ± 2 .

The OLED panel should light up under the rated current. After stabilization, the test should last for 60 minutes.

Heat and fire resistance

Heat resistance: The OLED panel should have sufficient heat resistance. The outer insulator material should have the function of contact protection and heat resistance.

The external insulating material parts that provide contact protection and the insulating material parts that hold the current-carrying parts in place should have sufficient heat resistance. For OLED panel plastic parts with additional insulation function, ball pressure test is not necessarily required.

Fire resistance: The insulating material parts that fix the current-carrying parts in place and the external insulating material parts that provide electric shock protection should be resistant to combustion and ignition. For materials other than ceramics, use the test in Chapter 13 of IEC60598-1 to check their eligibility.

Photo biological safety

The optical radiation of OLED products is not expected to reach the level required to mark the danger of ultraviolet, infrared or blue light, so no measurement is required.

GB/T 39237-2020 LED night scene lighting application technical requirements

- General requirements

LED lamps should meet the requirements of safety, reliability, advanced technology, reasonable economy, energy saving, environmental protection, and convenient maintenance.

LED lamps should be able to work normally within an ambient temperature of -40 ^50 . Special places should meet other requirements such as ambient temperature, humidity and corrosiveness of the specific place of use.

The driving power of LED lamps should meet the requirements of the national CCC certification.

LED lighting products for night scene lighting should meet the requirements of JGJ/T 163.

The control system should have safety, reliability, openness and scalability.

The maintenance factor of Led lamps should be selected according to Table 1.

Table 1 Led lamp maintenance factor

Lamp protection grade	Maintenance factor
ip65	0.7
ip65	0.65

- Specification classification requirements

Lamp classification

LED lamps for night scene lighting should be divided into LED lamps for functional lighting and LED lamps for landscape lighting.

LED lamps for functional lighting should be divided into LED mid-pole lighting (lamps), LED courtyard lamps, LED low-level lamps, and LED underground lamps according to the installation height or installation method of the lamps. Among them, LED low-level lamps can be divided into LED guardrail lamps and LED lawn lamps. , LED wall lamps and LED step lamps.

LED lamps for landscape lighting can be divided into LED project-light lamps, LED direct-view lamps and LED projection lamps according to the lighting method. Among them, LED project-light lamps can be divided into narrow beam, medium beam and wide beam according to the light distribution (see Appendix A) , LED direct-view lighting fixtures can be divided into LED point light source fixtures, LED line light source fixtures, LED surface light source fixtures and LED special-shaped fixtures according to their shape.

LED lamps for night scene lighting can be divided into white light LED lamps and colored light LED lamps according to the color table. Among them, white light LED lamps can be divided into non-adjustable color temperature and color temperature adjustable lamps.

Note: Color light LED lamps contain single or multiple color light channels, such as RGB, RGBW or RGBA.

LED lamps for night scene lighting can be divided into DC-powered lamps and AC-powered lamps according to the characteristics of the power supply.

- ØLamp specifications

The specifications of LED lamps for functional lighting should be classified according to Table 1 according to the rated luminous flux.

Table 1 Specification classification of LED lamps for functional lighting

Lamp name	Installation height/m	Rated luminous flux/lm	Maximum power/W
LED pole lighting (lamps)	6~15	5400	60
		7000	
		80	
		9000	
		100	
		14000	
LED courtyard light	2.5~6	160	25
		19000	
		220	
		25000	
		280	
		32000	
		360	
LED courtyard light	2.5~6	1700	25
		2100	
		30	
LED courtyard light	2.5~6	2800	40
		40	
LED courtyard light	2.5~6	3500	50
		50	

		0~1.2	700	12
LED low light	LED guardrail lighting			
900		15		
1100		18		
	0~1.2	100	6	
LED lawn light				
200		10		
270		12		
	0~1.2	180	3	
LED step lamps				
360		6		
500		8		
	0~1.2	700	12	
LED wall light				
900		15		
1100		18		
		0	15	1
LED buried	LED point light source			
Lamps				
80		3		
300		6		
LED linear light source				
600		12		
1000		18		
1500		24		

Direct-view lighting LED luminaire specifications should be classified according to Table 2 according to the rated luminous flux.

Table 2 Direct-view lighting LED luminaire specifications

Lamp name	Rated luminous flux/lm	Maximum power/W
LED point light source	25	0.5
	50	1
	100	2
	150	3
	220	4
	500	8
	800	12
	1000	16
LED linear light source	1500	24
	300	4
	600	8
	900	12
	1350	18
LED surface light source	1800	24
	2700	36
	900	12
1350	18	

1800	24
2700	36

The specifications of colored light LED lamps for landscape lighting should be classified according to Table 3 and Table 4 according to the rated luminous flux.

Table 3 Color light (RGB) LED lamps specification classification for landscape lighting

Lamp name	Rated luminous flux/lm	Maximum power/W
LED point light source	6	0.5
	25	1
	75	3
	240	8
	360	12
	480	16
	720	24
LED linear light source	120	4
	240	8
	360	12
	540	18
	750	24
	1200	36
LED surface light source	360	12
	540	18

750	24
1200	36
LED flood light	903
240	8
360	12
540	18
720	24
1080	3
1440	48
2100	72
3500	100
7000	200
14000	400
21000	600
28000	800
35000	1000

Table 4 Color light (RGBW) LED lamps specification classification for landscape lighting

Lamp name	Rated luminous flux/lm	Maximum power/W
LED point light source	6	0.5
40	1	
120	3	
320	8	

500		12	
700		16	
1000		24	
	180		4
LED linear light source			
350		8	
500		12	
750		18	
1000		24	
1500		36	
	500		12
LED surface light source			
750		18	
1000		24	
1500		36	
	120		36
LED flood light			
320		8	
480		12	
700		18	
960		24	
1440		36	
1920		48	
2880		72	
44500		100	

9000	200
18000	400
27000	600
36000	800
45000	1000

- Drive power

The driving power should be divided into constant current driving power and constant voltage driving power according to the output type.

The driving power should be classified according to the following output power ratings: 25W and below, 30W, 50W, 75W, 100W, 150W, 200W, 240W, 320W, 480W, 600W, 3kW, 6kW, 15kW.

Constant current drive power should be classified according to the following output DC current ratings: 350mA, 500mA, 700mA, 1050mA, 1400mA, 2100mA, 2800mA, 4200mA, 5600mA, 8000mA.

The rated value of the constant voltage drive power supply voltage should be classified according to Table 5.

Power type	Specification requirements
	12V 24V 36V 48V 240V
Fixed voltage type	
Centralized adjustable voltage type	Adjustable in the range of 48V~1000V

The size and weight specifications of the drive power supply should meet the requirements of Table 6.

Table 6 Drive power supply size and weight

Power/W		Size/mm		Maximum weight/kg	
length		width		height	
75W	200	220	80	45	1.5
240W	320	250	100	45	2

Dimmable drive power should be divided into current-adjustable type and current-duty-cycle type.

- Lighting control system

ED night scene lighting control system can be divided into analog control system and digital control system according to the control mode.

LED night scene lighting control systems should be classified according to Table 7 according to system functions and control methods.

Table 7 Classification of LED night scene lighting control system

category	System functions		Control method/protocol	Features	Use function
switch	Simple switch	Realize the lighting system is fully on and off	switch	Simple system structure and low cost; single control method	Can illuminate, landscape lighting for simple decoration
Group switch	Switch combination to achieve scene switching	Programming switch	0~10V	Simple system structure; low cost; simple control method	Functional lighting, landscape lighting for simple decoration
Dimming	Analog dimming	Realize dimming and scene switching through analog signal		Smooth and uniform dimming; poor reliability of voltage signal; not suitable for long-distance dimming	Functional lighting, decorative landscape lighting
number	Low speed	Realize dimming and scene low-speed switching	LoRa, NB-IoT,	Low power consumption; wide coverage; small data bandwidth; low transfer rate and low data load	Functional lighting, decorative landscape lighting
word	Switch	through digital signal	Power line carrier		
Tune					
Light					

	DMX512	
high speed	Realize rapid change of artistic effect through digital signal	Fast transmission speed;Landscape lighting for high refresh rate; low de-lighting and performance lay; strong anti-interference ability; high cost
Switch	KiNET	

	DMX512	
Achieve artistic effects through digital signals	RDM ORBIT	Landscape lighting for performances

- Lamp performance requirements

basic requirement

The maintenance factor of LED lamps should not be lower than the limit specified in Table 6.

Table 6 LED lighting maintenance factor

level	Maintenance factor
IP65	0.7
IP65	0.65

- Lamp safety requirements

The safety of LED lamps for night scene lighting should meet the requirements of GB 7000.1 and the following requirements:

- LED courtyard lamps should meet the requirements of GB7000.203;
- LED guardrail lamps and LED lawn lamps should meet the requirements of GB 7000.201.
- LED buried lamps and LED step lamps should meet the requirements of GB 7000.213.
- LED wall lamps should meet the requirements of GB 7000.202.
- LED floodlight lamps and LED wall wash lamps should meet the requirements of GB7000.7.
- The lamps used in swimming pools and similar places should meet the requirements of GB 7000.218.

LED lamps installed in places accessible to personnel should adopt safety extra-low voltage power supply or protection measures against accidental electric shock.

The protection level of LED lamps for night scene lighting should meet the following requirements:

- a) The protection level of lamps installed outdoors should not be lower than IP54;
- b) The protection level of LED underground lights should not be lower than IP67;
- c) The protection level of LED lamps installed in places such as leisure pools (swimming pools) should meet: Zone 0 should not be lower than IPX8;

It should not be lower than IPX5 in zone 1; it should not be lower than IPX4 in zone 2. The area division of the paddling pool (swimming pool) shall comply with the provisions of Appendix B of this standard.

Lamps and their accessories should have fall prevention measures. For lighting equipment accessible by personnel, isolation protection measures should be taken when the surface temperature is higher than 70°C.

LED lamps that cannot be directly installed on the surface of combustible materials should be marked in accordance with the provisions of GB 7000.1.

The surge protector of AC-powered LED lamps should be set independently, the output value of voltage protection level U_p should be less than the anti-surge voltage of the control device, and should not be greater than 2kV, wiring should have anti-misconnection measures, and the lightning protection level should meet the following requirements :

- a) LED luminaires for outdoor lighting (including adapted lightning arresters) should be able to meet the experimental conditions of 6kV in differential mode and 10kV in common mode

GB/T 17626.5-2008 requirements for category b) products;

- b) LED luminaires for indoor lighting (including adapted lightning arresters) should be able to meet the experimental conditions of 1kV differential mode and 2kV common mode

GB/T 17626.5-2008 specifies the requirements for category b) products.

The disturbance voltage of AC-powered LED lamps should comply with the relevant regulations of GB 17743.

The harmonic current limit of AC-powered LED lamps should meet the relevant regulations of GB 17625.1. Among them, AC-powered LED lamps with an active power of less than 25W should meet the relevant regulations of GB 17625.1 for discharge lamps with an active power of less than 25W.

- Electrical requirements for lamps

The rated voltage of LED lamps should comply with the power supply voltage regulations.

LED lamps should be able to work normally within the range of 90%~110% of the rated voltage, and special places should meet the requirements of the place of use.

The difference between the input power of LED lamps and the rated value should not be greater than 10% of the rated value or 0.5W.

Under rated power conditions, the power factor of AC-powered LED lamps should not be lower than the limit specified in Table 7

Table 7 Power factor requirements of AC-powered LED lamps

	≤5W	>5W
Measured power		
Power factor	0.7	0.9

- Lighting requirements

The initial luminous flux of LED lamps for night scene lighting should not be lower than 90% of the rated luminous flux, and should not be higher than 120% of the rated luminous flux.

When LED lamps for night scene lighting work within ±10% of the rated input voltage, the light output change should be within 5%.

After the installation of LED luminaires for functional lighting, the upward luminous flux ratio shall not exceed the limit specified in Table 8.

Table 8 The upper limit of the luminous flux ratio of LED lamps for functional lighting

	Environmental zone				
	E0 Area	E1	E2	E3	E4
Upstream luminous flux ratio (%)					
0	2.5%		5%		15%

Note: The division of environmental areas should be determined in accordance with Appendix C.

- Distribution.

The light distribution design of LED lamps used in window sill lighting should consider the control of spilled light.

The uniformity of the surface brightness of the LED surface light source should not be less than 0.8, and the calculation method should meet the requirements of Appendix D.

The surface brightness and viewing distance of LED lamps for direct-view lighting should be determined according to the design brightness of the building surface, the environmental brightness of the building surface and the installation method, and should be determined in accordance with Appendix E.

The average brightness of LED lamps for landscape signs and advertising lighting should not be greater than the limit specified in Table 9.

Table 9 Average brightness limit of LED lamps for landscape signs and advertising lighting

	Environmental zone				
	E0 :zone	E1 :	E2 :	E3 :	E4 :
	d0.1	d50	d400	d800	d1000
Average brightness /					
(Cd/m2)					

The glare limit of LED lamps for functional lighting should not be greater than the limit specified in Table 10.

Table 10 Glare limits of LED lamps for functional lighting

Lamp name	Installation height/m	cd/1000lm	
		Maximum light intensity	
LED pole light	80°±90°	±90°	
	6~15	150	30
LED courtyard light	2~6	200	50
	0~1.2	/	1-0
LED low light	LED guardrail lighting		
LED lawn light			
LED step lamps			
LED wall light			
LED Buried Light	LED point light source	0	/ 10

LED linear light source

Note: The table shows the luminous intensity in all directions at the specified angle ²formed by the downward vertical axis of the lamp after it is installed.

The luminous intensity of LED lamps for direct-view lighting should not be greater than the limit specified in Table 11.

Table 11 Luminous intensity limits of LED lamps for direct-view lighting

Lighting technical parameters		Application conditions			Environmental zone	
E0 zone	E1	E2	E3	E4		
	0	2500	7500	10000	25000	
Luminous inten- sity/cd	Before lights out					
	0	0	500	1000	2500	
Lights out period						

The ratio of peak luminous intensity to luminous flux (K value) of LED floodlights should meet the requirements of Table 12.

Table 12 The relationship between the beam angle and K value of LED floodlights

	4	4	8	12	15	22	30	45
Beam angle								
	e120	e80	e22	e15	e10	e3.5	e2.5	e1.5
cd/lm								
Peak light intensity/lumi- nous flux (K value)								

è

Note: The beam angle of LED floodlights is calculated according to 50% of the maximum light intensity, and the two directions (0° -180°, 90° -270°) in the light intensity distribution of the lamp are taken as the minimum value of the corresponding beam angle.

Chromaticity requirements of lamps

The rated correlated color temperature of LED lamps for functional lighting should not be higher than 5000K.

Generally, the color rendering index of LED lamps for functional lighting should not be less than 60.

The deviation of the chromaticity coordinates of the white LED lamps measured from the nominal color temperature and the chromaticity coordinates of the CIE 1976 uniform chromaticity scale specified in Chapter 4 of GB/T 7921-2008 should not be greater than 0.0055; the coordinate deviation of the LED wall washer It should not be greater than 0.0033. The chromaticity coordinates of the nominal color temperature can be selected according to Appendix F.

The spatial chromaticity uniformity of LED lamps for night scene lighting should meet the following requirements:

- a) The deviation of the chromaticity coordinates of white light LED lamps in different directions and their weighted average values shall not exceed 0.007 in the CIE 1976 uniform chromaticity scale chart specified in Chapter 4 of GB/T 7921-2008;
- b) The deviation of the chromaticity coordinates of the LED flood light in different directions and its weighted average is in the 17CIE1976 uniform chromaticity scale chart specified in Chapter 4 of GB/T7921-2008, and should not be greater than 0.004.

The deviation of the chromaticity coordinate and the initial value during the life cycle of LED lamps for night scene lighting shall meet the following requirements:

- a) The deviation of the chromaticity coordinates during the life cycle of white LED lamps from the initial value should not be greater than 0.012 in the CIE1976 uniform chromaticity scale chart specified in Chapter 4 of GB/T 7921-2008.
- b) The deviation of the chromaticity coordinates during the life cycle of the LED flood light from the initial value should not be greater than 0.007 in the CIE1976 uniform chromaticity scale chart specified in Chapter 4 of GB/T7921-2008.

The color consistency between LED linear light source lamps and LED surface light source lamps should meet the requirements of Appendix E.

The dominant wavelength range and color purity of LED lamps for night scene lighting should meet the requirements of Table 13.

Table 13 The dominant wavelength range and color purity requirements of LED lamps for night scene lighting

colour	Red light	Green light	Blue light	Yellow light
	610~700	508~550	455~475	585~600
Dominant wavelength range				
	e94	e72	e90	e93
Color purity limit				

- Lamp dimming requirements

The dynamic range of dimming LED lamps for night scene lighting should meet the following requirements:

a) The lower limit of the dynamic range of dimming LED lamps for functional lighting should not be less than 10% of the rated luminous flux;

b) The dynamic range of LED luminaire dimming for landscape lighting should be 0%~100%.

The dimming output characteristics of LED lamps should meet the following requirements:

a) LED lamps for functional lighting should adopt linear output of luminous flux;

b) LED lamps for landscape lighting should determine the ³correction factor according to the ambient brightness.

Note: ³correction refers to the function transformation of the dimming output curve.

The LED lamp for functional lighting should be equipped with a device for constant illumination control.

The LED lights for landscape lighting used for performances should adopt the control mode of DMX512-1990 standard protocol.

The dimming performance of LED lamps within the set dimming range should meet the following requirements:

a) When dimming, the deviation between the measured luminous flux of the lamp and the set value should not exceed 5%;

b) The frequency of the current pulse should not be less than 200Hz for the driving power supply that adopts the current-adjusting duty ratio control method for dimming.

- ØEnergy efficiency requirements of lamps

The luminaire efficacy of white LED courtyard luminaires should not be lower than the limits specified in Table 14.

Table 14 Luminaire efficacy limits of white LED courtyard luminaires

60		Correlated color temperature			
		Color rendering index			
3000K		3000K~4000K		4000K	
communicate with	Direct current	communicate with	Direct current	communicate with	Direct current
80	90	85	95	90	100

The luminaire efficacy of white LED lawn luminaires, LED step luminaires, and LED recessed wall luminaires shall not be lower than the limits specified in Table 15.

Table 15 Luminaire efficacy limits of white LED lawn lamps, LED step lamps, and LED recessed wall lamps

project		White LED Correlated Color Temperature					
3000K		3000K~4000K		4000K			
communicate with	Direct current	communicate with	Direct current	communicate with	Direct current		
	60		60		65		70
					75		80
Color rendering index							85
80	55	60	65	70	75	80	
90	50	55	60	65	70	75	

The luminaire efficacy of white LED floodlights should not be lower than the limits specified in Table 16.

Table 16 Luminaire efficacy limits of white LED floodlights

(Color rendering index e80)		White LED Correlated Color Temperature					
3000K		3000K~4000K		4000K			
communicate with	Direct current	communicate with	Direct current	communicate with	Direct current		
	B 50% 4°	25	30	35	40	45	50
beam angle							
4°B 50% 18°	55	75	70	85	80	95	
18°B 10% 46°	55	75	70	85	80	95	
46°B 10% 70°	60	80	75	90	85	100	

70°	B 10%	100°	64	85	79	95	89	105
B10%	100°	77	90	85	100	94	110	

Note: B 10% refers to the beam angle calculated according to the 10% maximum light intensity value, and B50% refers to the beam angle calculated according to the 10% maximum light intensity value.

The luminaire efficacy of white LED lamps for direct-view lighting should not be lower than the limit specified in Table 17.

Table 17 Luminaire efficacy limits of white LED lamps for direct-view lighting

	3000	3000~4000	4000
Correlated color temperature/K			
	80	85	90
Lamp performance limit			

The luminaire efficacy of LED luminaires for multi-channel direct-view lighting shall not be lower than the limits specified in Table 18.

Table 18 Luminaire efficacy limits of LED lamps for multi-channel direct-view lighting

colour	Lamp performance
R	40
G	45
B	25
RGB	35
W 1800K-2700K	70
W(3000K-6500K)	75
RGBW 1800K-2700K	45
RGBW(3000K-6500K)	50
Y	55

The luminaire efficacy of multi-channel LED lawn lamps and LED step lamps should not be lower than the limits specified in Table 19.

Table 19 Luminaire efficacy limits of multi-channel LED lawn lamps and LED step lamps

Power supply type		Lamp color	
R		G	B
communicate with	70	70	70
	80	80	80
Direct current			

The luminaire efficacy of multi-channel LED floodlights should not be lower than the limits specified in Table 20.

Table 20 Luminaire efficacy limits of multi-channel LED floodlights, in lumens per watt

		colour					
project		R		G		B	
communicate with	Direct current	communicate with	Direct current	communicate with	Direct current	communicate with	Direct current
B 50%	4°	21	25	35	40	12	13
beam							
angle							
B 50%	18°	45	48	70	80	22	32
B 10%	46°	45	48	70	80	22	32
B 10%	70°	45	48	70	80	22	32
B 10%	100°	45	48	70	80	22	32
B 10%	100°	50	55	80	90	25	29

Note: B 10% refers to the beam angle calculated according to the 10% maximum light intensity value, and B 50% refers to the beam angle calculated according to the 10% maximum light intensity value.

- Lamp durability requirements

The maintenance rate of the luminous flux of the LED lamps during the normal operation of 3000h should not be less than 96%, and the maintenance rate of the luminous flux of the 6000h should not be less than 92%.

The life of LED lamps for night scene lighting should not be less than 25000h.

The damage rate of LED lamps after one year of normal operation should not be higher than 0.5%.

- Drive power requirements

basic requirements

The driving power supply should meet the relevant regulations of GB 19510.1 and GB 19510.14.

The driving power supply should be able to work normally at a temperature of -40°C to 55°C and a relative humidity of 10% to 100%. Special places should meet other requirements such as ambient temperature, humidity and corrosiveness of the specific place of use.

The drive power supply should adopt a standard interface and should be easy to install, maintain and replace.

- Electrical performance

The power of the driving power supply should match the power of the lamp.

The output current setting value should be 70%~100% of the rated value when the drive power supply is working normally.

The disturbance characteristics of the power terminals of the driving power supply for LED lighting should meet the relevant regulations of GB/T 9254 and GB/T 17743.

The performance of the driving power supply for LED lighting should not be changed under the condition that the fluctuation of the rated input voltage value of $\pm 10\%$ and the total harmonic distortion rate of the AC input voltage is not more than $\pm 5\%$.

The harmonic current limit of the driving power supply for LED lighting should comply with the provisions of Article 7 of GB 17625.1-2012, and its total harmonic distortion rate under the rated voltage should not exceed the limit specified in Table 21.

Table 21 Limits of total harmonic distortion rate

Power range/W	Load ratio/%	Total harmonic distortion rate /%
5Pd75	100	15
75	20	
50	25	
P 75	100	10

75	15
50	20

Note: This standard refers to the 2nd to 40th harmonic current components.

The rated frequency of the AC input voltage of the driving power supply for LED lighting should be 50Hz, and it should be able to work normally within the fluctuation range of the rated input frequency ± 3 Hz.

The power factor and efficiency of the drive power supply should not be lower than the limits specified in Table 22.

Table 22 Drive power factor and efficiency limit

Power range/W	Load ratio/%		Power factor		effectiveness/%
Isolated			Non-isolated		
Pd5	100	0.8	80		86
5Pd75	100	0.92	85		89
75	0.9		83		86
50	0.9		80		83
75Pd200	100	0.96	88		92
75	0.94		85		89
50	0.9		83		87
P 200	100	0.96	90		93
75	0.94		88		90
50	0.9		85		88

The starting output voltage or current overshoot amplitude of the drive power supply should not exceed 10% of the rated voltage or current value.

The deviation of the output current of the constant current drive power supply should not exceed 3% of the nominal value; the deviation of the output voltage of the constant voltage drive power supply should not exceed 5% of the nominal value.

When LED lighting is used for dimming with a driving power supply, the deviation between the measured effective value of the output current of the power supply and the set value should not exceed 5%.

When the driving power supply integrates the function of the lighting controller, it shall have a control interface for docking with the lighting control system.

The temperature coefficient of the drive power supply should not exceed $\pm 0.1\%$.

- Safety requirements

The driving power supply for LED lighting should be able to automatically output an alarm signal when the internal temperature reaches the protection setting value, and should be able to automatically reduce the power or directly cut off the power for protection.

The protection level of the driving power supply for LED lighting shall not be lower than the requirements of Table 23.

Table 23 Requirements for protection level of driving power supply for LED lighting

Power type	Protection level
Built-in	IP20
	IP43
	IP65
Independent (including centralized powerindoor supply)	
outdoor	

The lightning protection performance of the driving power supply for LED lighting should meet the following requirements:

- a) The driving power supply for outdoor lighting (including the adapted lightning protection device) should be able to meet the requirements of category b) products specified in GB/T 17626.5-2008 under the experimental conditions of 6kV in differential mode and 10kV in common mode;
- b) Under the experimental conditions of differential mode 1kV and common mode 2kV, the driving power supply for indoor lighting (including the adapted lightning protection device) should meet the requirements of category b) products specified in GB/T 17626.5-2008.

- Durability

When the maximum temperature point of the housing does not exceed 75 , the life of the driving power supply for LED lighting should not be less than 50000h.

Under the working conditions specified in 7.1.2, the drive power failure rate for LED lighting should not exceed 0.5%.

The product should be able to work normally after a 500-hour durability test at a high temperature (85).

Control system requirements

- Basic requirements

The control device should be easy to replace and repair on site.

The control system should have the hardware and software interface with the heavy current system, and should be able to realize the linkage control of the two systems.

The control system should be able to control the lighting fixtures individually, grouped or zoned according to functional requirements.

- Control requirements

The control system should be able to achieve control functions through data collection and analysis, automatic function presets, etc., and should meet the following requirements:

- a) It should be able to realize switch control in accordance with the stated requirements;
- b) For places where dimming is required, the light illumination (luminance) should be adjusted according to the set value;
- c) In places where scene switching is required, automatic or manual switching of the set scene mode should be possible according to the stated requirements, and on-site adjustments should be possible.

The control system should be able to set, modify and reset system parameters locally or remotely.

The control system should have an information collection function, which can display and record lighting system related information, and can automatically generate analysis and statistical reports.

The control system should reserve the linkage interface with other systems.

The control system can have environmental information and human-computer interaction functions.

The master control platform shall have complete operation authority restrictions.

The control system should be able to preset a variety of lighting scene control schemes according to requirements, and should meet the following requirements:

- a) It should be able to perform marshalling control on actual effect requirements, and should be able to display its working status;
- b) It should be able to display the operating status of the power supply;
- c) When the power distribution and control system fails, it should automatically send out sound and light alarm signals;
- d) Direct manual control should be set;
- e) It should support remote query and monitoring through mobile devices.

The LED in-situ lighting control system should determine the network structure of the control system according to the effect and field conditions.

The LED control system of the media facade used for video playback should adopt wired control.

LED lamps for functional lighting can be equipped with or externally connected to an automatic control device that reduces the illuminance at night. LED courtyard lights should be equipped with a single light control device.

LED lamps for functional lighting should be equipped with smart sensors or an external sensor control interface, which can automatically turn off the lights or reduce the illuminance level according to the needs of use.

LED floodlights and LED linear lamps used in areas where no one stays for a long time should be equipped with smart sensors or an external sensor control interface, which can automatically turn off the lights or reduce the illumination level according to the needs of use.

The control system refresh frequency should not be lower than 25 frames per second, and the control system communication delay should not exceed 30ms.

For the media facade lighting around the residential area and in the restricted area, the playing time of each frame of the picture shall not be less than 2s, and the switching time shall not be less than 1s.

The landscape lighting should be set according to the operating time of the night mode, weekday mode, holiday mode and major holiday mode, and should meet the following requirements:

- a) Normal mode time should be within 4h after sunset, and its brightness limit should be determined according to Table 9;
- b) The mid-night mode should determine the brightness limit level according to the area to which it belongs, and should not exceed the E2 zone standard.

Mandatory or voluntary standards

Are there some mandatory or voluntary standards applicable ? If yes, which ones ?

Please indicate the reference of the applicable standards

Regard to your products, if you want to check what are the technical requirement /toxicology/design... shall respect, thanks to check the labelling of your product first and find code of standards in labelling.

Step 1:

Enter this link: https://docs.google.com/spreadsheets/d/1BDqhm609YySgDqs8cPOQUd-Dc_ZYPEEt7v-lv1sWB0CA/edit#gid=1640363751

Step 2:

Enter the code of product standard selected by you in PACE and open the link in red.

Enter the code of mandatory safety standard above.

Step3:

Open link and find documents of standard, your products shall respect all requirements in file.

PS: If you can't find the translation file in link above, please do not hesitate to contact with CHINA Conformity Leader by mail "chinadmilegal@decathlon.net".

Physical and mechanicals properties

What are the physical and mechanicals properties required ?

GB 30255-2019 limited energy efficiency value and energy efficiency level of LED products for indoor lighting

The level of energy efficiency

LED products for indoor lighting have three energy efficiency levels, among which level 1 has the highest energy efficiency.

The lighting effect of each grade of LED tube shall not be lower than the provisions in table 1.

Rated power/W	CCT) K	Photosynthetic efficiency		
		lm/ W		
Level 1	Level 2	Level 3		
	CCT<3500	95	80	60
d5				
CCTe3500	100	85		65
5	CCT<3 500	105	90	70
CCTe3500	110	95		75

The directional integrated LED lighting effect of each grade shall not be lower than that specified in table 2.

Table 2 Directional integrated LED light energy efficiency rating

Lamp type	CCT)		Photosynthetic efficiency	
	K		lm/ W	
Level 1	Level 1		Level 1	
	CCT<3500	95	80	65
PAR16/ PAR20				
CCTe3500	100	85	70	

	CCT<3 500	100	85	70
PAR30/ PAR38				
CCTe3500	105	90	75	

GB/T 39008-2020 double-ended integrated LED lamp performance requirements

Product classification and model

Product Categories

Luminous flux specifications are classified according to the nominal luminous flux of the lamp, corresponding to the tube diameter and length of the lamp, as shown in Table 1.

Table 1 Luminous flux specifications of lamps

Nominal pipe diameter	Nominal length mm	Luminous flux specification lm Rated luminous flux	
16	550	600	600
800		800	
900		900	
1050		1050	
850	900		900
1200		1200	
1150	1300		1300
1600		1600	
1850		1850	
2100		2100	
3900		3900	
1450	1600		1600
2000		2000	

3900		3900	
26	600	800	800
900		900	
1000		1000	
900	1200		1200
1500		1500	
1200	1600		1600
1800		1800	
2000		2000	
2500		2500	
1500	2000		2000
2200		2200	
2500		2500	
2700		2700	
3700		3700	

Light distribution type

According to the light distribution type of the lamp, it is divided into two light distribution types: wide beam type (code: W) and narrow beam type (code: N).

Hue specifications

According to the color temperature of the lamp, it is divided into 2700K (incandescent lamp color), 3000K (warm white), 3500K (white), 4000K (cool white), 5000K (neutral white) and 6500K (daylight) and other standard colors. , Its hue code and color coordinate target value are shown in Table 2.

Table 2 Color specifications

Serial number	Hue specifications	Hue code	Color coordinate target value
x		y	

1	6500K (Daylight)	65	0.313	0.337
2	5000K (Neutral white light)	50	0.346	0.359
3	4000K (Cold white)	40	0.380	0.380
4	3500K (white)	35	0.409	0.394
5	3000K (warm white)	30	0.440	0.403
6	2700K (Incandescent color)	27	0.463	0.420
P27		0.458		0.410

The manufacturer can manufacture lamps of non-standard colors according to the requirements of users, but should also provide the target value and tolerance range of the chromaticity coordinates of the non-standard colors.

P27 represents the hue code whose color coordinate is closest to the Planck curve and the color temperature is 2700K.

ØModel indicates that the model of the regular lamp consists of five parts: the first part indicates the code of the lamp (IPT stands for double-ended integrated LED lamp); the second part indicates the luminous flux specification of the lamp; the third part indicates the light distribution type of the lamp, including width Beam type (code: W, can be omitted) and narrow beam type (code: N); the fourth part represents the color characteristics of the lamp, using color rendering index code (generally the code for color rendering index not less than 80 is "8 ", the code not less than 90 is "9") and the color code (CCT divided by 100). For example, 865 represents the color rendering index of 80 and the color temperature of 6500K; the fifth part represents the lamp cap model and nominal length And other information, if the above two or more kinds of content appear at the same time, separated by ".", the producer can choose whether to standardize this part.

skills requirement

Applicable power supply voltage and ambient temperature of the lamp The lamp should be able to work normally at 90%~110% of the rated power supply voltage and within the range of the nominal power supply voltage. The lamp should be able to work normally within the rated ambient temperature range.

Lamp dimensions and lamp holder

The external dimensions of the lamp should conform to the values specified in the parameter table of the corresponding specification double-capped fluorescent lamp in GB/T1682 that the manufacturer or seller claims to replace. The size of the lamp cap on the finished lamp should meet the requirements in GB/T1406.2.

Electrical parameters

Power: When the lamp is working under rated voltage and rated frequency, the actual power consumption does not exceed 10% of the rated power.

Displacement coefficient

When the lamp works under rated voltage and rated frequency, its actual displacement coefficient should not be lower than the nominal value by more than 0.05, and should not be lower than the requirements in Table 3.

Table 3 Modification coefficient requirements

Rated power P	Pd2W	2w pd5w	5w pd25w	p 25w
		e0.4	e0.7	e0.9
Becomes a coefficient	no limit			

Luminosity parameters

Luminous flux: The rated luminous flux of the lamp should meet the requirements of Table 1. The initial value of the luminous flux of the lamp should not be lower than 90% of the rated luminous flux and not higher than 120% of the rated luminous flux.

Initial luminous efficiency: The initial luminous efficiency of the lamp shall not be lower than the regulations in Table 4.

Table 4 The initial light effect of the lamp

level	65/50/40	35/30/27/P27
	Hue code	Hue code
	120	110
CE	105	95
	90	80

Luminous flux angle

The beam angle of the lamp should conform to the light distribution type stated by the anufacturer or seller.

Color parameters The chromaticity performance of the lamp should meet the requirements of Table 5, and the average value of the lamp chromaticity tolerance should not be greater than the value specified in Table 5.

The initial average value of the lamp's general color rendering index R should not be lower than the value specified in Table 5, and the measured value of each sample should not be lower than the value specified in Table 5 by more than 3 numerical values; R>0.

Table 5 Chromaticity performance of the lamp

Hue specifications							Hue code	Chromaticity parameter
SDCM							General color rendering index R	
Color coordinate target		Correlated color temperature target value	Chromaticity tolerance KSDCM					
x						y		
6500K	65	80	0.313	0.337	6430	5		
(Day-light)								
5000K			50		0.346		0.359	5000
(Neutral white)								
4000K			40		0.380		0.380	4040
(Cold white)								
3500K			35		0.409		0.394	3450
(white)								
3000K			30		0.440		0.403	2940
(warm white)								
2700K			27		0.463		0.420	2720
(Incandescent color)								
P27			0.458			0.410		2700

The manufacturer can manufacture lamps of non-standard colors according to the requirements of users, but should also provide the target value and tolerance range of the chromaticity coordinates of the non-standard colors.

The target value of the correlated color temperature is a non-assessment item.

life

Average life: The average life of the lamp should not be less than 15000h.

Early failure: The lamp should not appear early failure during the aging process.

Luminous flux maintenance rate The luminous flux maintenance rate of the lamp at 3000h and 6000h should not be lower than the specified value in Table 6.

Table 6 Lumen maintenance rate of the lamp

number	Rated life	Luminous flux maintenance rate	Luminous flux maintenance rate
1	15000	93.1	86.7
2	20000	94.8	89.9
3	25000	95.8	91.8
4	30000	96.5	93.1
5	35000	97	94.1
6	40000	97.4	94.8
7	45000	97.7	95.4
8	50000	97.9	95.8

Note: If the rated life of the lamp is not the value listed in the table, the requirement of the luminous flux maintenance rate is calculated from the two adjacent values by linear interpolation

Electromagnetic compatibility characteristics

Electromagnetic disturbance Electromagnetic disturbance should meet the requirements of GB/T17743.

Harmonic current: Harmonic current should meet the requirements of GB17625.1.

Logo and packaging

Marking requirements: The minimum packaging of lamps and lamps or the marking content on the instruction manual should meet the requirements of Table 7.

Table 7 The minimum packaging of lamps and lamps or the marking requirements on the instruction manual

projec	Light body	The small
		est package or instruction manual of the lamp
product name	—	×
model	×	×
	×	—
Date of manufacture (year, quarter or month, year and month are represented by numbers, and season is represented by Roman characters; or production batches with traceable production date)		
Applicable power supply voltage and ambient temperature	×	×
Rated luminous flux/lm	—	×
Light effect	—	×
Light distribution type	—	×
rated power	×	×
Modification coefficient	—	×
General color rendering index	—	×

Correlated color temperature	—	×
Rated life	—	×
Lamp Type	—	×
Implementation standard name or number	—	×

Note: "×" is the required content; "—" is the selected content.

Electrical aspects

What are requirements regarding electrical aspects ?

Inflammability

Are there any specific requirements to ensure inflammability?

Hygiene

What are the rules regarding hygiene ?

Radioactivity

What are the rules regarding radioactivity ?

Radiofrequency (2G,3G,4G,5G, Bluetooth, Ant+, GPS, Wifi...)

What are the rules regarding the radiofrequency?

Products making noise

What are the requirements regarding products making noise ?

Chemicals substances

What are the requirements regarding chemicals substances ? Which ones are forbidden ? Limited ? Do we have to clearly indicate all substances that are in the product, or is the indication of major substances enough?"

HG/T 3318-2018 Natural rubber adhesive for fabric and leather

requirements

Technical index

The technical indicators should meet the requirements of Table 1.

Table 1 Technical indicators

project	index
look	Light yellow viscous liquid, no impurities, precipitation, gel
	12000~40000
Viscosity	
	2.00
Folding bonding strength	
	10
Non-volatile content	

Note: The flash point does not exceed 28 , which belongs to Class A hazardous chemicals

Harmful Substance

The limit of hazardous substances should meet the requirements of Table 2.

Table 2 Limits of hazardous substances

project	index
d	5
benzene	
d	150
Toluene	
d	150
N-hexane	
d	5

1.2 Dioxyethane

d 20

Total halogenated hydrocarbons

d 550

Total volatile organic compounds

Inspection rules

Product inspection is divided into factory inspection and type inspection.

Ex-factory inspection: appearance, viscosity, hem bond strength, non-volatile content.

Type inspection: when a new product is put into production; when there are major changes in materials, process and equipment, which may affect product performance; when normal production reaches 100 kettles; when production is stopped for more than 1 month and resumes production.

Packaging requirements

Can you please indicate if there are some requirements regarding the packaging of our products (mainly they are in paper, carton, plastic, textile) ? If yes, please explain them.

GB/T36949—2018 Double (Capped LED Lamps Designed to Retrofit Liner Fluorescent Lamps) Performance Specifications

3.9.2 Packaging requirements

The lights should be individually packaged and then packed in a box.

The number of lights in the box should be marked on the box.

The product should be accompanied by a product certificate or stamped with a stamp and stamped with the date of inspection.

Laboratory

In which laboratory can we perform the testing of the product according to the standard?

Sanctions / Fines

What are the sanctions / fines in case of non conformity ?

Comments

Comments

Certifications or registrations

Are there some certifications or registrations applicable to the product ?

The power supply voltage above 36 V and not more than 1000 V —Recessed luminaires ,Fixed general purpose luminaires, Movable general purpose luminaires are compulsorily required to be certified before being sold on the market.

Cosmetics Supervision and Administration Regulation

Laboratory

In case of mandatory certification/ registration that we must follow please precise the name and address of any laboratories where we can conduct them

Comments

Comments

Conformity documents

Documents

What are the conformity documents ? Can you please list them ?

Format

Does it exist a specific format for the conformity documents ? If yes, can you please provide us an example in english ?

Comments

Comments

Importation rules

General rules

What are the general rules in order to import the product ?

Regulatory requirements for electrical and electronic product

Regulatory requirements for product import

- The electrical and electronic products imported may not violate the compulsory standards or the standards that must be implemented as prescribed by laws, administrative regulations and rules, and shall comply with the national or industrial standards for the restricted use of the hazardous substances contained in electrical and electronic products.
- The entry-exit inspection and quarantine institutions shall conduct port verification and statutory inspection of the electrical and electronic products imported. The Customs shall verify the Customs Clearance Form for Goods Entering China issued by entry-exit inspection and quarantine institutions and undergo the customs clearance formalities under the relevant provisions.

Regulatory requirements for import of mechanical and electrical products

The regulatory requirements for the import of mechanical and electrical products are mainly stipulated in Administrative Measures for Import of Mechanical and Electrical Products and Implementation Measures of Automatic Import License for Mechanical and Electrical Products and other regulations. Since it has not been retrieved that your company sells products listed in the Catalog of Restricted Imports of Mechanical and Electrical Products and Electrical Products or the Catalog of Mechanical and Electrical Products Prohibited from Importing, this section is mainly aimed at the mechanical and electrical products listed in Automatic Import License Mechanical and Electrical Products Catalog, which are classified as product with automatic import license.

General regulatory requirements

- To import mechanical and electrical products with automatic import license, the import entity shall file an application with the Ministry of Commerce or its authorized institution for an Automatic Import License before it goes through the customs formalities.

Application requirements

- The import of a mechanical and electrical product in Automatic Import License Mechanical and Electrical Products Catalog shall conform to the provisions of the following laws and administrative regulations:

(1) The provisions of Standardization Law of the P.R.C. regarding the standards and enforcement standards on ensuring the human health, personal safety and property safety;

(2) The provisions of [the Environmental Protection Law of the People's Republic of China](#) and other laws and administrative regulations on environmental protection;

- (3) The provisions [the Law of the People's Republic of China on the Prevention and Control of Air Pollution](#) and other laws and administrative regulations;

(4) The relevant provisions of [the Regulation of the People's Republic of China on Certification and Accreditation](#);

(5) The import of measurement instruments shall conform to the relevant provisions of [Measurement Law of the P.R.C.](#) and Administrative Measures for Supervision of Imported Instruments of Measurement;

(6) The import of radio transmission equipment shall conform to Administrative Regulations for Radio of the P.R.C.;

(7) Conforming to relevant provisions of other laws and administrative regulations.

Application process

The applicant may submit a written application or an on-line application.

1.
 1. Written application procedure: The import applicant can obtain the "Mechanical and Electrical Products Import Application Form" from the issuing agency or download (copy) from the website authorized by the Ministry of Commerce, and submit it to the corresponding competent authority together with other relevant written materials.
 2. Online application procedure: The applicant for import logs on the website authorized by the Ministry of Commerce, enters the national mechanical and electrical product import document management system, and truthfully fills in the "Mechanical and Electrical Product Import Application Form" online as required and submits it to the corresponding competent authority

The procedure for issuing "Automatic Import License" for mechanical and electrical products is as follows:

1.
 1. For the import of an mechanical and electrical product that shall be handled by the local or departmental mechanical and electrical office, as soon as the local or departmental mechanical and electrical office receives a Form of Application for Import of mechanical and electrical Products and relevant materials, whose contents are correct and which meet the formal requirements, it shall issue an Automatic Import License. In a special case, it may issue an Automatic Import License within 10 working days.
 2. For the import of a mechanical and electrical product that shall be handled by the Ministry of Commerce, the local or departmental mechanical and electrical office shall, within 3 working days after it receives a complete set of application materials, submit to the Ministry of Commerce the electronic data thereof upon verification. As soon as the Ministry of Commerce receives the corresponding data, it shall issue an Automatic Import License. In special cases, it may issue an Automatic Import License within 10 working days.

Documents

What are the required documents ?

Time period

In which time period may we obtain the right to import ?

Administration

What is the administration in charge of importations ? Name and address ?

Comments

Comments

Inspection of the product

Inspection method

Does it exist an inspection procedure applicable to the product ? If yes, which administration is in charge of ? Can you please describe the procedure ?

Cosmetics Supervision and Administration Regulation

Commercialization rules

Notification / Declaration

What are the general rules in order to comercialize a product ?
Is there a notification / declaration to do ?

Regulatory requirements for electrical and electronic product

regulatory requirements for

product sales

The sellers of electrical and electronic products may not sell the electrical and electronic products that violate the national or industrial standards for the restricted use of the hazardous substances contained in electrical and electronic products.

Documents

What are the required documents ?

Delay

What is the delay to obtain the right to commercialize ?

Display in store

Is there some specific rules on the way to display the product in store ?

Display on internet

Is there some specific mentions to display the product on internet ?

Information to consumers

Do some specific information to have be given to consumers when the product is placed on the market ?

Comments

Comments

Administration

What is the administration in charge of commercialization ? Name and address ?

Documents